WATER RESOURCES AND LAND SUBSIDENCE

LESSON 8



OBJECTIVES

At the end of the lesson, students should be able to:

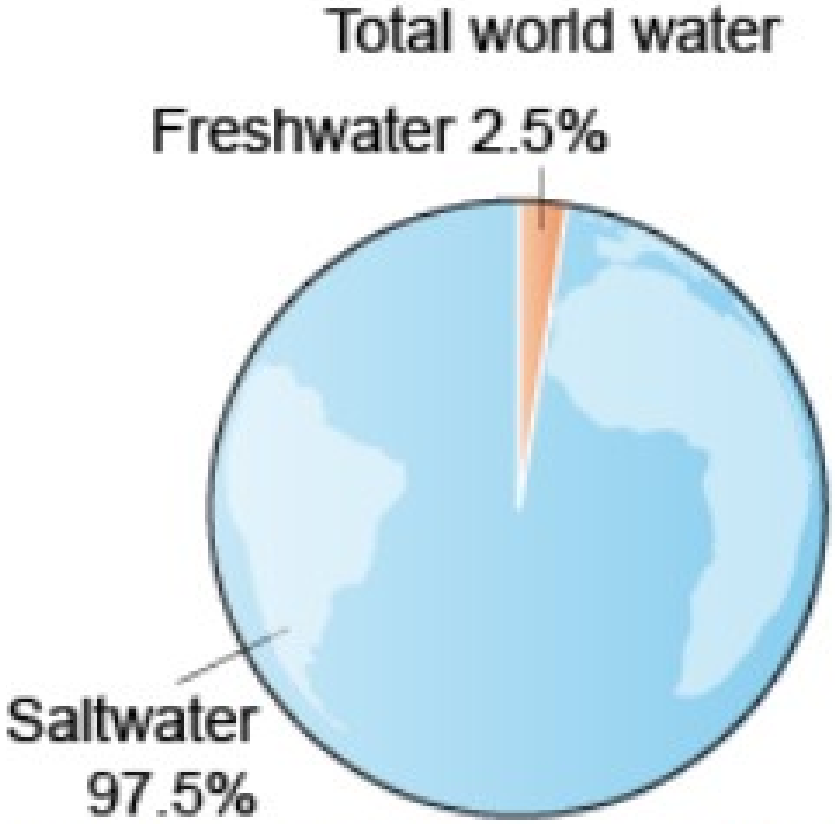
1. **Identify** the major types and sources of water resources.
2. **Explain** common water resource issues and their effects on people and the environment.
3. **Describe** land subsidence, its causes, effects, and relationship with groundwater extraction.
4. **Discuss** sustainable strategies for managing water resources and preventing further land subsidence.

DISTRIBUTION OF EARTH'S WATER

97.5% saltwater (oceans)

2.5% freshwater

- ~69% glaciers
- ~30% groundwater
- ~1% surface water



TYPES OF WATER USE

- **Agricultural (70%)** – irrigation for crops.
- **Industrial (20%)** – manufacturing, energy production.
- **Domestic (10%)** – household use (drinking, cleaning, sanitation).

WATER RESOURCES

Water resources are sources of water that are useful or potentially useful to humans, ecosystems, agriculture, industry, and other activities.



Water Resource	Description	Examples	Major Uses
1. Surface water	Water found on Earth's surface.	Rivers, lakes, reservoirs, streams.	Drinking water, irrigation, industry.
2. Groundwater	Water stored beneath Earth's surface in soil and rocks.	Aquifers, wells	Drinking water, irrigation, industry.
3. Rainwater	Water collected from rainfall.	Rainwater harvesting systems.	Household use, irrigation, gardening.

Water Resource	Description	Examples	Major Uses
4. Glaciers and ice caps	Frozen freshwater stored as ice.	Mountain glaciers, polar ice.	Important freshwater storage
5. Wetlands	Areas where water covers or saturates the soil.	Marshes, swamps, mangroves.	Flood control, water filtration, habitat.
6. Seawater	Saltwater found in oceans and seas.	Pacific Ocean, Philippine Sea.	Fisheries, transport, desalination.
7. Atmospheric water	Water present in the atmosphere as vapor or clouds.	Water vapor, clouds.	Part of the water cycle.

Water Resource Issue		Possible Effects
1. Water scarcity	Available water is insufficient to meet demand.	Shortage of drinking and irrigation water.
2. Water pollution	Harmful substances enter water bodies.	Diseases, ecosystem damage.
3. Over-extraction	Water is removed faster than it is naturally replenished.	Declining groundwater levels.
4. Drought	Long period of unusually low rainfall.	Agricultural losses and water shortages.
5. Flooding	Excess water covers normally dry land.	Property damage, erosion, displacement.

WATER SCARCITY

- **Water scarcity** occurs when the available water supply is insufficient to meet the needs of people, agriculture, industry, and ecosystems.

1. Physical water scarcity	There is **not enough water available naturally.**	A region experiences prolonged drought.
2. Economic water scarcity	Water may exist, but people **lack adequate infrastructure or resources to access it.**	A community has a water source but lacks treatment or distribution systems.

WATER POLLUTION

Water pollution occurs when harmful substances enter rivers, lakes, groundwater, oceans, or other water bodies.

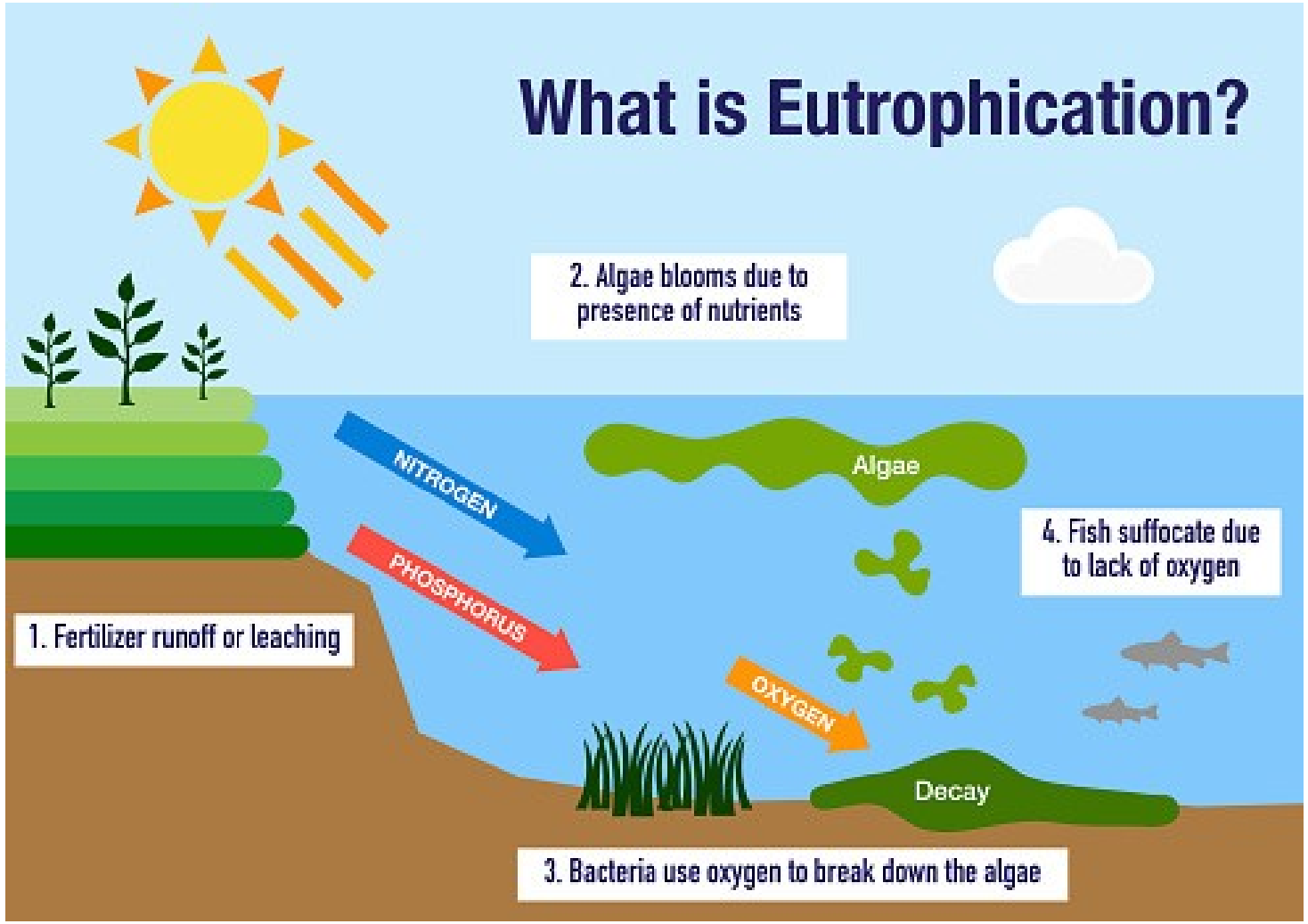


COMMON SOURCES OF WATER POLLUTION

Source	Examples	Possible Effects
Domestic wastewater	Sewage, detergents, household waste.	Disease and oxygen depletion.
Agriculture	Fertilizers, pesticides, animal waste.	Algal blooms and contamination.
Industry	Chemicals, heavy metals, wastewater.	Toxic effects on humans and aquatic organisms.
Solid waste	Plastics, garbage.	Blocked waterways and harm to wildlife.
Mining	Sediments and chemical contaminants.	Water contamination.
Oil pollution	Oil spills and petroleum products.	Damage to marine ecosystems.

EXAMPLE: NUTRIENT POLLUTION

- Excess fertilizers containing **nitrogen and phosphorus** can enter waterways.
- **Fertilizer runoff → excess nutrients → rapid algae growth → decomposition → oxygen decreases → fish and aquatic organisms may die**
- This process is called **eutrophication**.


What is Eutrophication?
2. Algae blooms due to presence of nutrients
NITROGEN
PHOSPHORUS
Algae
4. Fish suffocate due to lack of oxygen
1. Fertilizer runoff or leaching
OXYGEN
Decay
3. Bacteria use oxygen to break down the algae

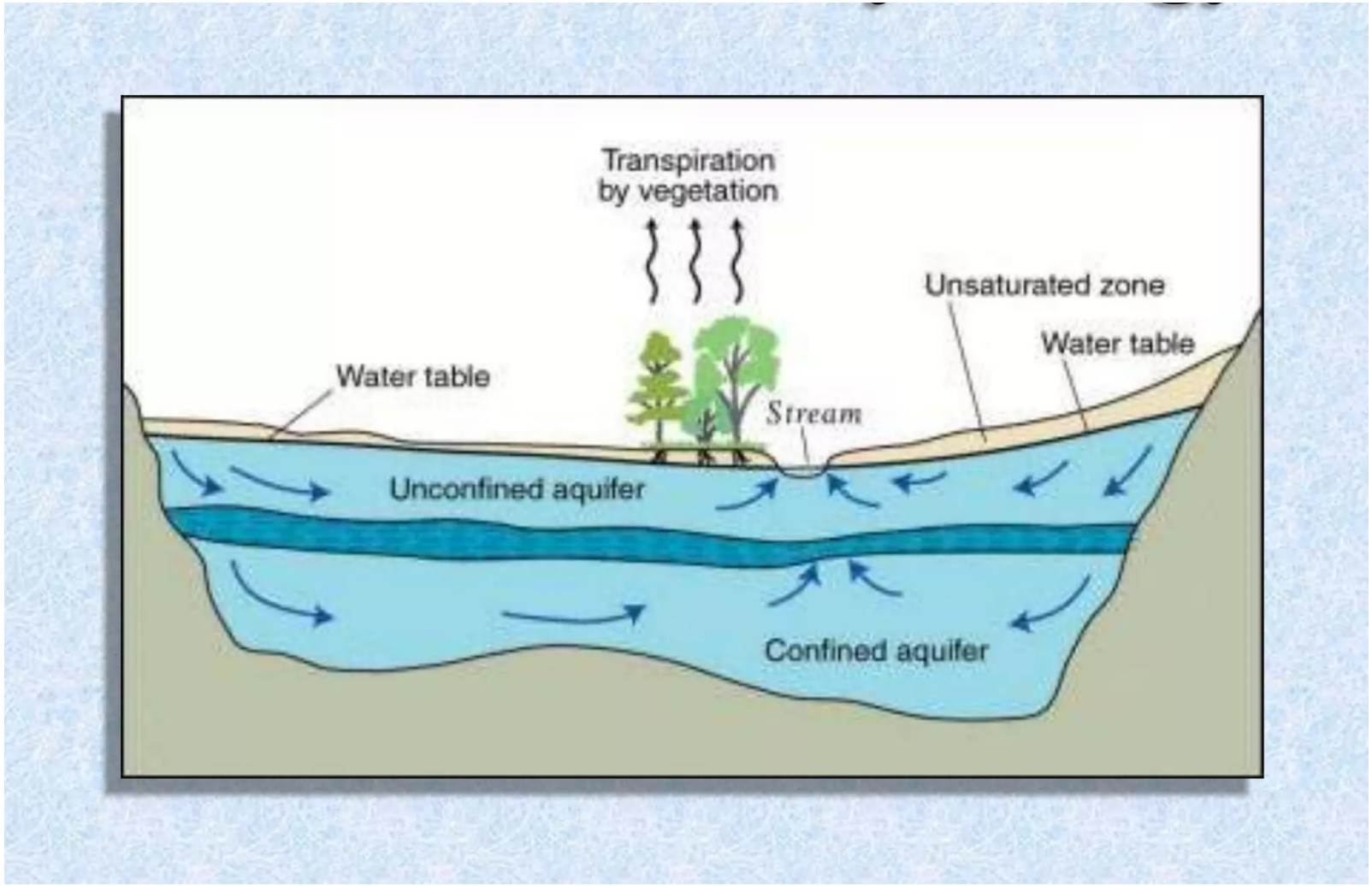
Water Resource Issue		Possible Effects
6.Saltwater intrusion	Seawater enters freshwater aquifers.	Groundwater becomes salty.
7. Deforestation	Removal of vegetation reduces infiltration and affects watersheds.	Increased runoff and erosion.
8. Climate change	Changes rainfall patterns, temperature, and extreme events.	Droughts, floods, changing water availability.
9. Urbanization	Development increases paved surfaces.	Reduced groundwater recharge and increased runoff
10. Agricultural demand	Large amounts of water may be needed for irrigation.	Increased pressure on rivers and aquifers.

SALTWATER INTRUSION

- **Saltwater intrusion** occurs when seawater moves into freshwater groundwater systems, particularly in coastal areas.
- Normally, freshwater in a coastal aquifer helps prevent seawater from moving inland. However, **excessive groundwater pumping** can lower the groundwater level.

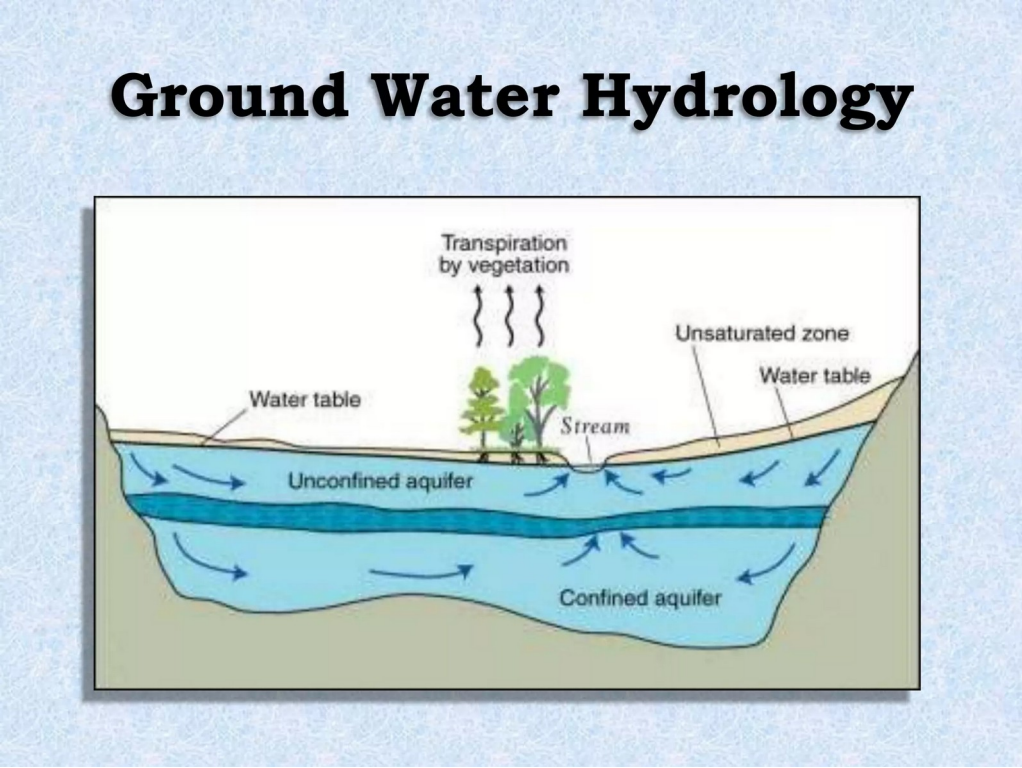
Process

- **Excessive groundwater pumping → groundwater level decreases → pressure of freshwater decreases → seawater moves inland → groundwater becomes salty**



UNDERSTANDIN G GROUNDWATER

GROUNDWATER



Groundwater is water stored below Earth's surface in the spaces and cracks within soil, sand, gravel, and rocks.

- **Water table**: The upper level of the saturated groundwater zone.
- An **aquifer** is a layer of permeable rock, sediment, or soil that can **store and transmit groundwater**.
  - **Unconfined** - *open, shallow, easily affected*
  - **Confined** - *covered, deep, under pressure*

GROUNDWATER RECHARGE

- **Groundwater recharge** is the process by which water moves from the surface into the soil and eventually reaches underground aquifers, replenishing groundwater supplies. It helps maintain water availability, prevents groundwater depletion, and reduces the risk of land subsidence.

Methods:

A. **Natural** Groundwater Recharge

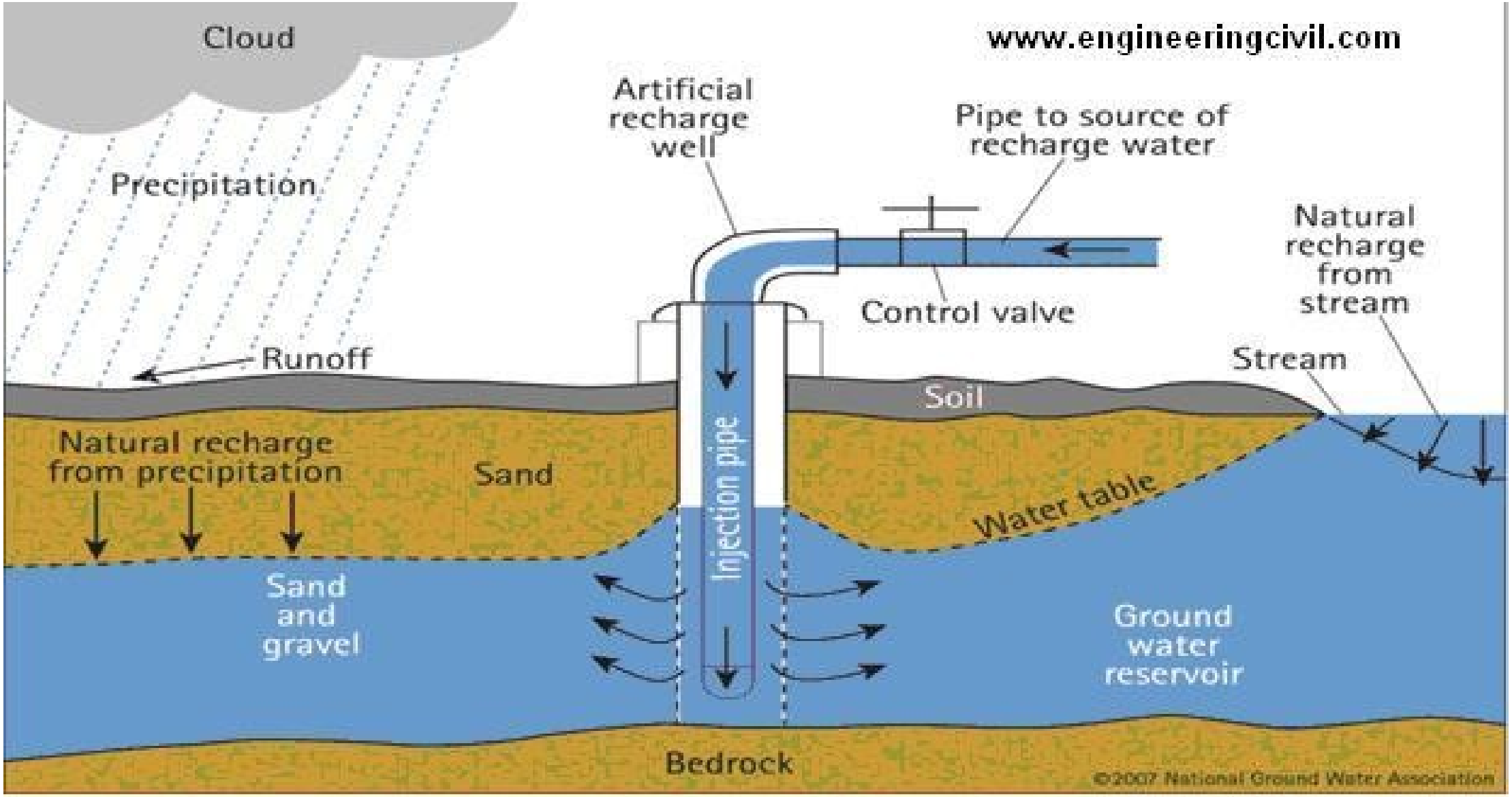
B. **Artificial or Managed** Groundwater Recharge

A. **NATURAL** GROUNDWATER RECHARGE

	Explanation	Example
1. Rainfall infiltration	Rainwater enters the soil and moves downward toward the groundwater system.	Rain falling on an open field.
2. River and stream infiltration	Water from rivers and streams seeps through their beds into the ground.	A river replenishing a nearby aquifer

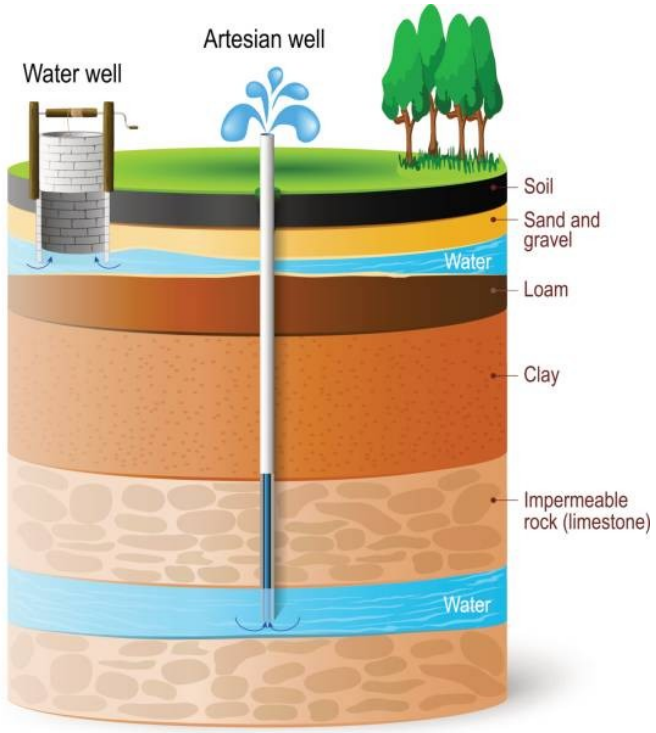
B. **ARTIFICIAL OR MANAGED** GROUNDWATER RECHARGE

Method	Explanation	Example
Recharge ponds/basins	Water is temporarily stored in shallow basins so it can slowly infiltrate into the ground.	A basin filled with collected rainwater
Percolation ponds	Ponds are specifically designed to allow water to seep downward into groundwater systems.	A community recharge pond
Recharge wells	Water is intentionally introduced into an aquifer through a properly designed well.	Treated water injected into a suitable aquifer
Rainwater harvesting	Rainwater is collected and directed to areas where it can infiltrate or be stored for later use.	Roof runoff directed toward a recharge garden
Managed aquifer recharge (MAR)	Water is intentionally managed and directed into groundwater storage for later use.	Capturing excess water during the rainy season for aquifer recharge


Cloud
www.engineeringcivil.com
Artificial recharge well
Pipe to source of recharge water
Precipitation
Natural recharge from stream
Control valve
Runoff
Stream
Soil
Natural recharge from precipitation
Sand
Injection pipe
Water table
Sand and gravel
Ground water reservoir
Bedrock
©2007 National Ground Water Association

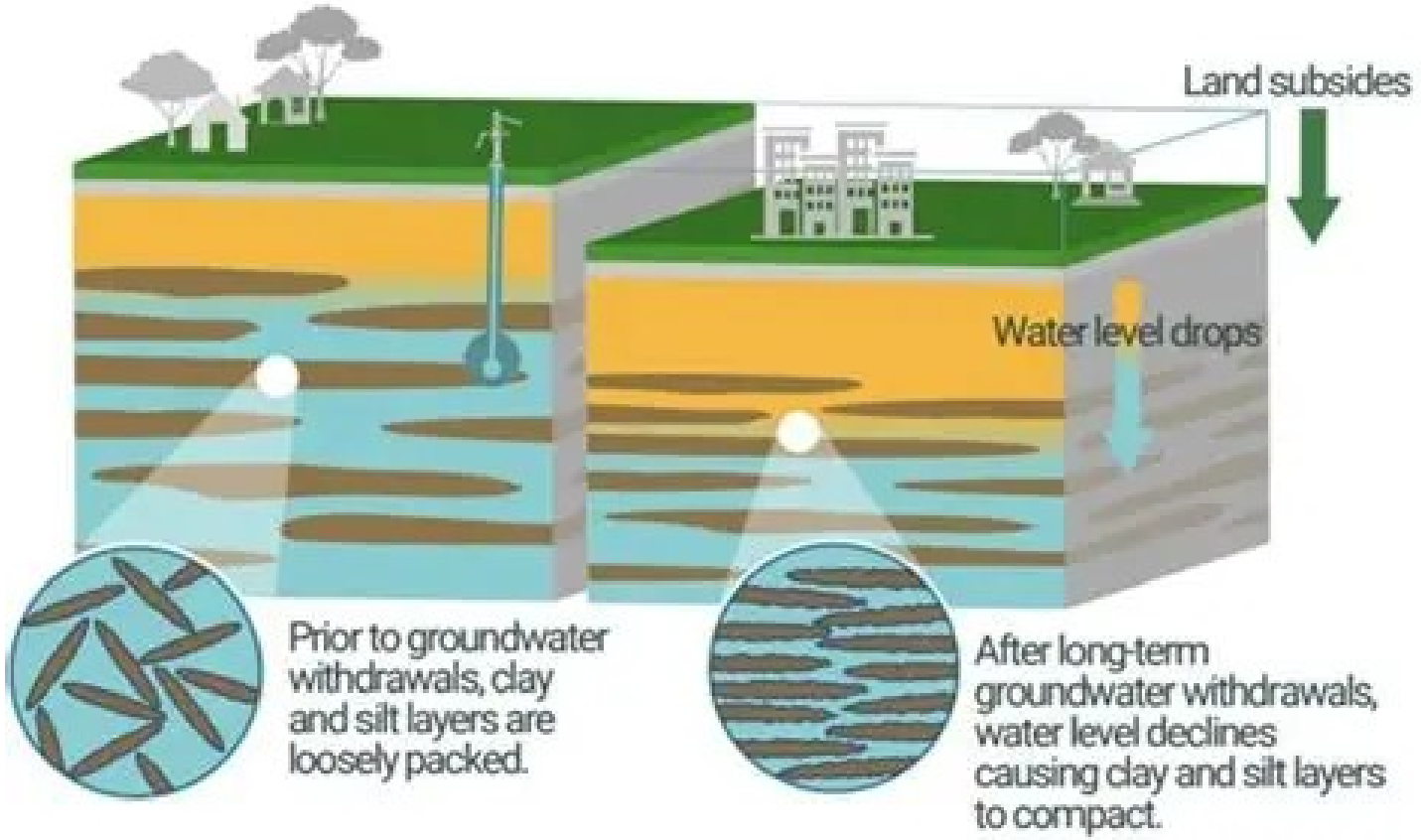
GROUNDWATER EXTRACTION

- **Groundwater extraction**: Removal of groundwater through wells or pumping





LAND SUBSIDENCE

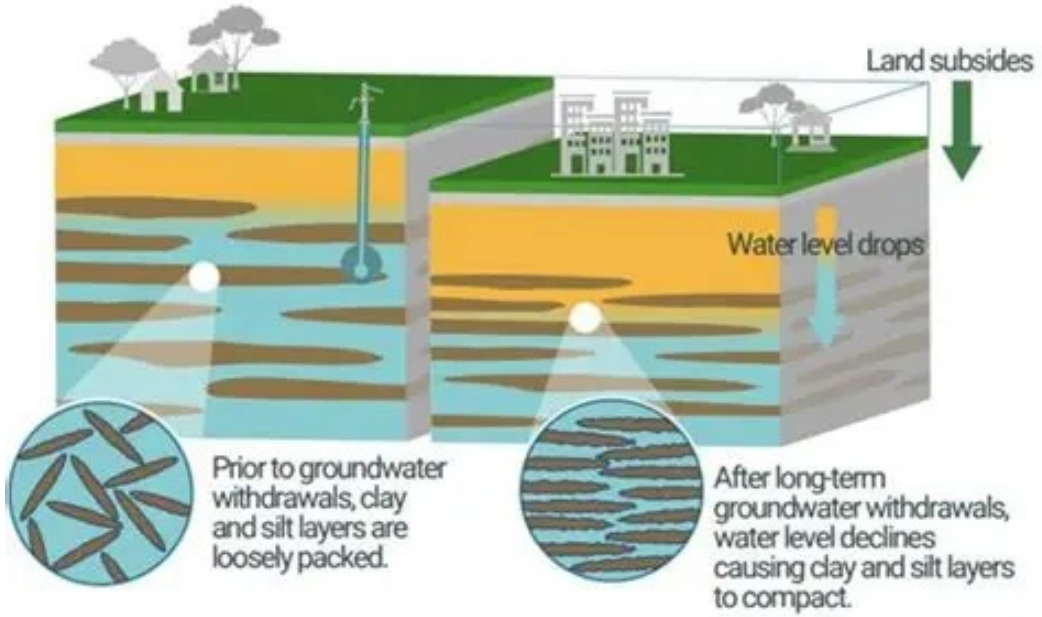


LAND SUBSIDENCE

- **Land subsidence** is the **gradual sinking or lowering of the Earth's land surface**.
- It can occur **natu**rally, but **human activities** can greatly **accelerate it.**

- **Excessive groundwater extraction → groundwater level decreases → underground sediments compact → land surface sinks**

HOW DOES GROUNDWATER EXTRACTION CAUSE LAND SUBSIDENCE?

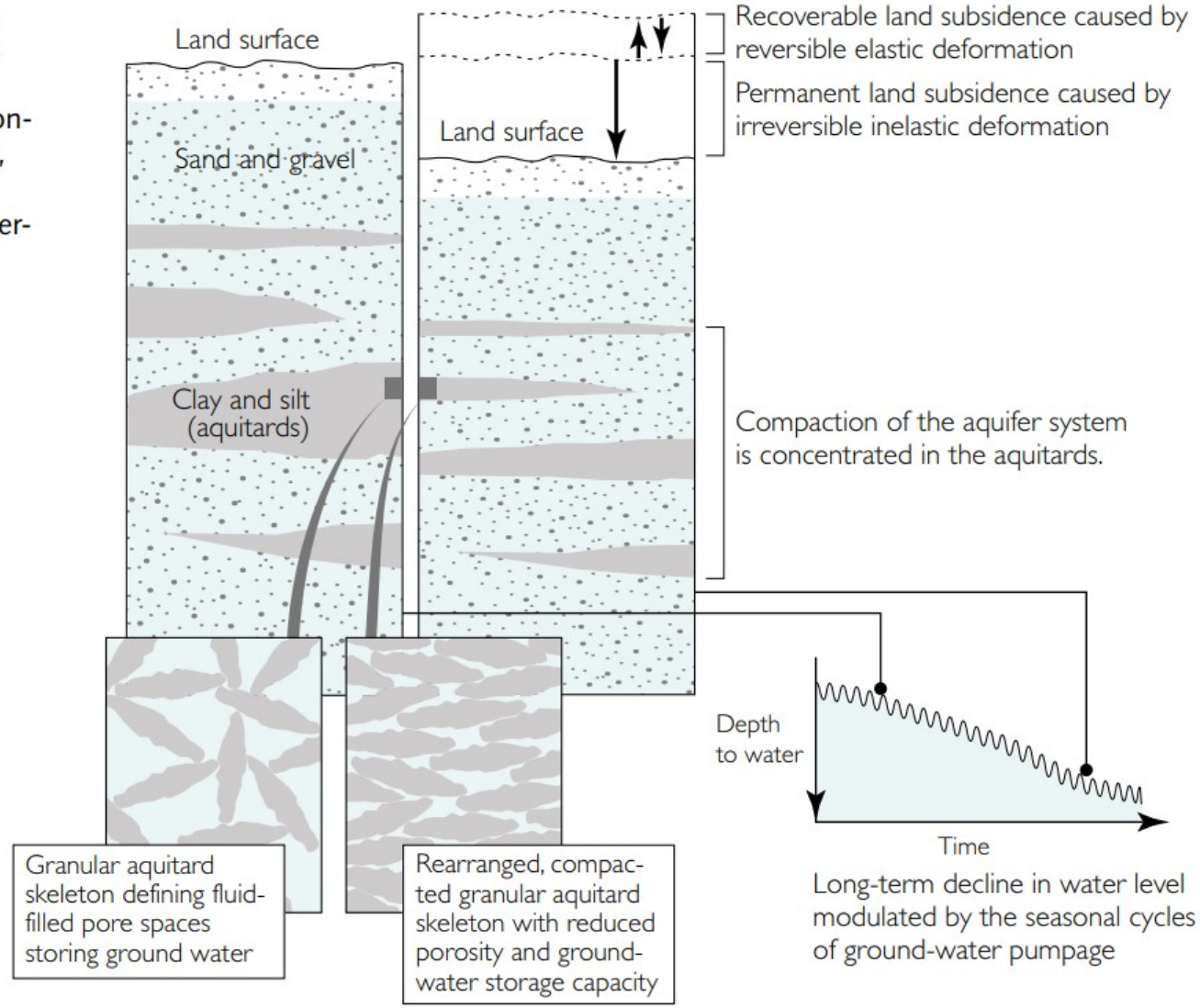


- Groundwater is often stored in spaces between particles of clay, sand, and other sediments.
- When groundwater is **pumped out**, the **water pressure** that helps support these sediments **decreases.**
- The sediments may become **compressed or compacted.**

STEP-BY-STEP PROCESS

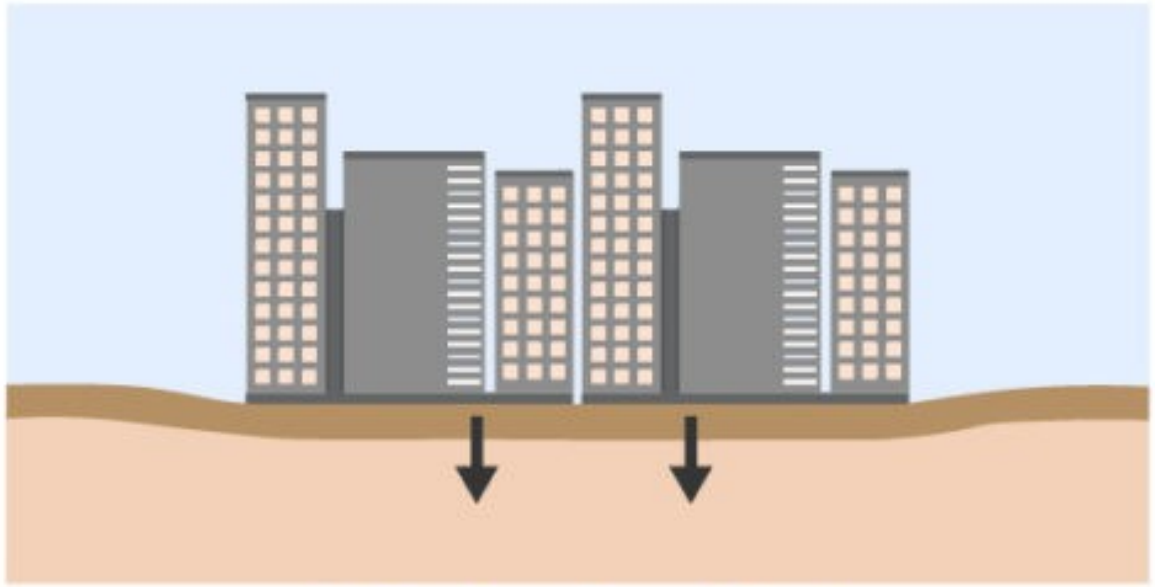
Step	What Happens?
1. Groundwater is stored underground.	Water fills spaces between sediment particles.
2. Large amounts of groundwater are pumped.	The groundwater level decreases.
3. Water pressure decreases.	Sediment particles become more closely packed.
4. Sediments compact.	The underground layer becomes thinner.
5. Land surface moves downward.	The ground gradually sinks.
6. **Land subsidence**	Buildings, roads, drainage systems, and

When long-term pumping lowers ground-water levels and raises stresses on the aquitards beyond the preconsolidation-stress thresholds, the aquitards compact and the land surface subsides permanently.



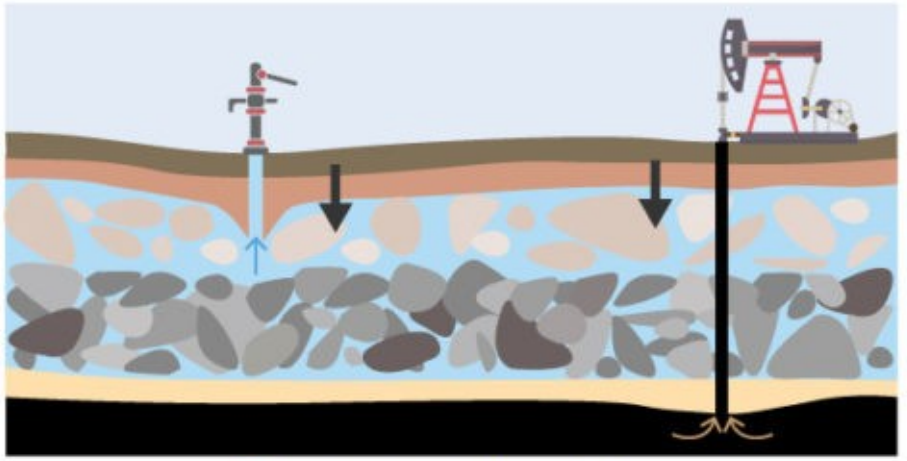
OTHER CAUSES OF LAND SUBSIDENCE

Cause	Explanation
Oil and gas extraction	Removal of underground fluids can reduce pressure and cause ground movement
Mining	Removal of underground materials can cause the ground to collapse or settle
Natural sediment compaction	lose sediments may naturally become compressed over time
Construction and loading	Heavy buildings and infrastructure can compress weak soils
Tectonic processes	Movement of Earth's crust can cause changes in land elevation



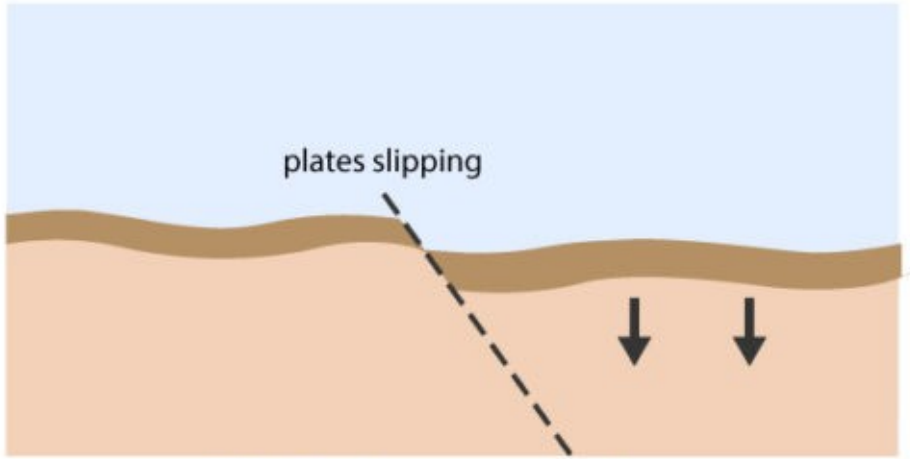
Infrastructural load

the weight of buildings causes compaction



Resource extraction

e.g. groundwater, oil and gas



Tectonic movements

e.g. earthquakes

EFFECTS OF LAND SUBSIDENCE

Effect	Explanation
Flooding	Lower land is more easily flooded during heavy rainfall.
Coastal flooding	Subsidence can increase the relative effect of sea-level rise.
Damage to buildings	Uneven sinking can cause cracks and structural problems.
Road damage	Roads may crack, deform, or become uneven.

EFFECTS OF LAND SUBSIDENCE

Effect	Explanation
Drainage problems	Gravity-based drainage becomes less effective.
Damage to pipelines	Water, sewage, and gas pipelines may be stressed.
Saltwater intrusion	Lower land and altered groundwater conditions can increase coastal vulnerability.
Loss of agricultural land	Farms may become waterlogged or increasingly exposed to flooding.
Economic losses	Communities may spend more on repairs and flood protection.

LAND SUBSIDENCE AND CLIMATE CHANGE

- Land subsidence and climate change can **increase each other's impacts** in vulnerable areas.

For example:

- **Groundwater extraction → land subsidence**

At the same time:

- **Climate change → sea-level rise**

When both occur in a coastal area:

- **Land sinks + sea level rises → greater relative flooding risk**
- This is particularly important for **low-lying coastal communities and agricultural areas**.

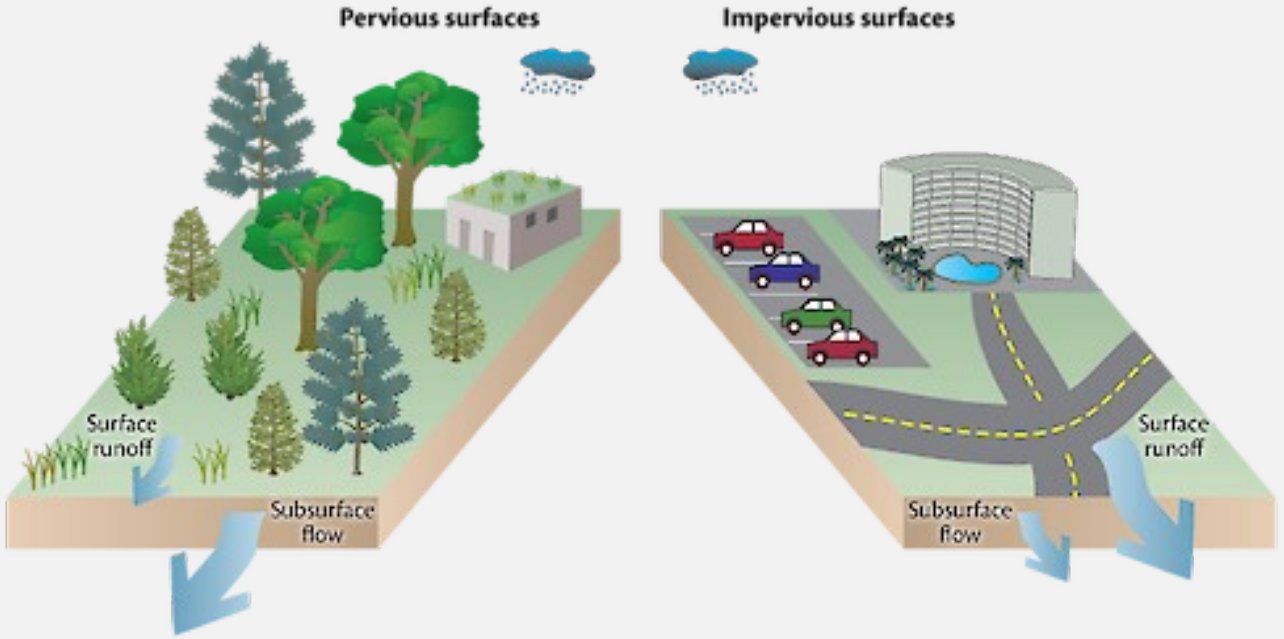
HOW CAN WE HELP PREVENT LAND SUBSIDENCE?

- Reduce excessive groundwater pumping
- Use surface water responsibly
- Harvest rainwater
- Improve irrigation efficiency
- Protect recharge areas

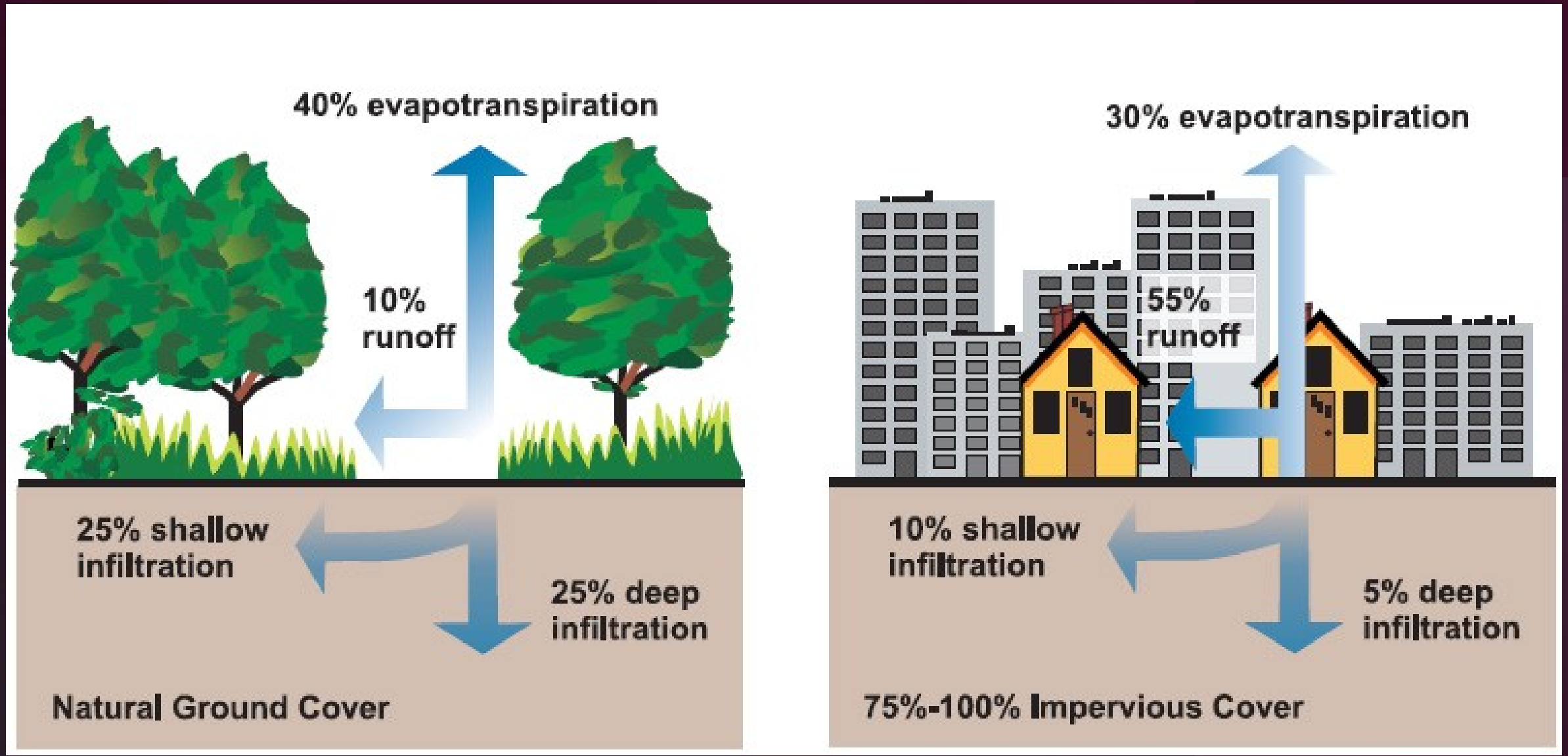
- Increase green spaces
- Use **permeable** pavements
- Monitor groundwater levels
- Regulate deep wells
- Reuse treated wastewater
- Protect watersheds

IMPERVIOUS MATERIALS

- **Impervious materials** are surfaces that **do not allow water to infiltrate** into the ground.
- This increases the risk of:
  - **Urban flooding**
  - **Flash floods**, especially during heavy rainfall
- Common examples: **concrete, asphalt, rooftops, and paved roads**.



Pervious surfaces such as grass, soils, and 'green roofs' allow water to infiltrate the ground, slowing and reducing runoff and recharging groundwater. Impervious surfaces such as cement, asphalt, and roofing prevent infiltration, increasing the volume and velocity of surface runoff which carries nutrients and sediments with it. Diagram courtesy of the Integration and Application Network (ian.umces.edu). University of Maryland Center for Environmental Science. Source: Dennison, W.C., J.E. Thomas, C.J. Cain, T.J.B. Carruthers, M.R. Hall, R.V. Jesian, C.E. Wazniak, and D.E. Wilson. 2009. *Shifting Sands: Environmental and cultural change in Maryland's Coastal Bays*. IAN Press, University of Maryland Center for Enviromental Science.


40% evapotranspiration
10% runoff
25% shallow infiltration
25% deep infiltration
Natural Ground Cover
30% evapotranspiration
55% runoff
10% shallow infiltration
5% deep infiltration
75%-100% Impervious Cover

THE PHILIPPINE CLEAN WATER ACT

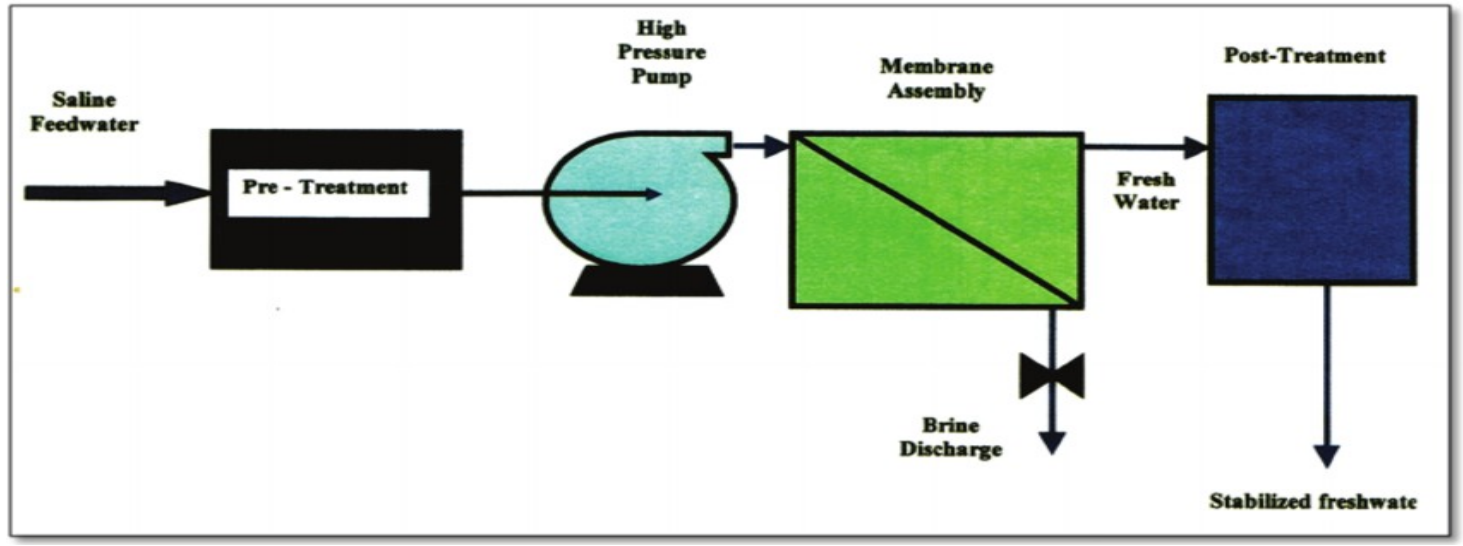


- The **Philippine Clean Water Act (Republic Act No. 9275)**, enacted in **2004**, is the country's primary law for **protecting water bodies from pollution** and **ensuring sustainable water resource management.**
- Despite its strong legal framework, implementation gaps persist.



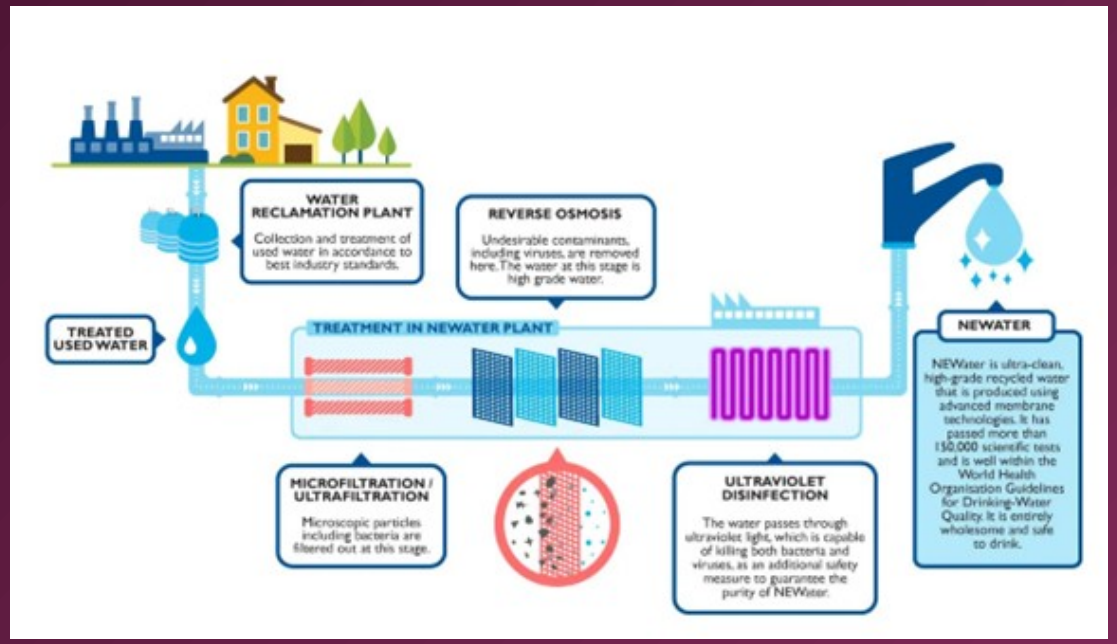
SUSTAINABLE WATER MANAGEMENT STRATEGIES

1. **Conservation**
   - Fixing leaks, efficient fixtures, and mindful use.
2. **Protecting Water Sources**
   - Preventing pollution and overuse of rivers, lakes, and aquifers.
3. **Efficient Irrigation**
   - Using drip or sprinkler irrigation instead of flood irrigation.
4. **Community Action**
   - **Rainwater harvesting**, protecting watersheds.
5. **Technology**
   - **Desalination, wastewater recycling** (Singapore's NEWater).





- **Desalination** is the process of **removing salts** and minerals from seawater or brackish water to produce freshwater for human use.





WASTEWATER RECYCLING – SINGAPORE'S NEWATER

- **NEWater** is Singapore's ultra-clean recycled water made from treated wastewater using microfiltration, reverse osmosis, and ultraviolet disinfection.
- It is used for **industrial purposes**, **reservoir topping (**adding treated water to a reservoir to increase or maintain its water level**),** and even drinking after blending, providing a sustainable and reliable water source.